Progress Report and Next Steps

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1 Executive Summary

What has been done since my proposal defense: no actual results or work for the benchmarking project were part of my proposal defense, so every tangible part of the benchmarking project is new. In brief, I have done the following:

1. **gRNA assignment comparison:** `SCEPTRE`, `pertpy`, `CLEANSER`, and `crispat` have been evaluated and compared on both simulated and real datasets.
   (a) I developed a nextflow pipeline to reproducibly parallelize gRNA assignment over dataset-method pairs, with metrics tracking, and GPU usage for `pertpy`
   (b) I created an original mechanism for simulating gRNA matrices with covariate effects and known ground truth
   (c) I compared and contrasted all 4 of these methods on 2 simulated datasets and 2 real datasets, both statistically and computationally, as well as running the `sceptre-pipeline` on both real datasets
2. **Association analysis comparison:** so far, `SCEPTRE`, `Mixscale`, and `FR-Perturb` have been evaluated and compared on real datasets for on-target performance
   (a) I created on-target downsamples of two real datasets, and developed a nextflow pipeline to reproducibly compare and contrast all 3 of these methods statistically using pre-determined guide assignments

Major steps remaining:

1. Association: negative control analysis
2. Association: computational benchmarking
3. Possibly adding more datasets and methods
4. Writing the paper, thesis, and thesis defense

2 Detailed Report

I will now give a review of the key findings so far.

2.1 Guide Assignment

For this project, we are currently comparing 4 methods on 2 simulated datasets, and 2 real datasets.

The methods are as follows:

1. `pertpy`'s `assign_mixture_model()`: a Poisson-Gaussian mixture model fit via MCMC to the log-transformed counts.
2. `CLEANSER`: either a Poisson-NB or NB-NB mixture, fit to the raw counts via MCMC, with a scaling factor for total library size.
3. `crispat`'s `ga_poisson_gauss()`: a Poisson-Gaussian mixture fit to the log-transformed counts. MAP estimates are obtained via stochastic variational inference.
4. `SCEPTRE`: a latent variable Poisson GLM, fit via a modified EM algorithm, controlling for some guide covariates and possibly batch

The datasets are as follows:

1. Simulated: two datasets with 100k cells were generated, where each guide is an independent sample of the same process. Covariate effects learned from Replogle-RD7 were included. The fraction of perturbed cells was set to either 1.5% or 0.5%. Currently only a large perturbation effect size is used.
2. Real: Replogle-RD7 (possibly downsampled to the first 2500 guides and first 50,000 cells) and Gasperini (also possibly downsampled to the first 2500 guides and first 50,000 cells)

2.1.1 Simulation results

Figure 1 gives a comparison of the real and simulated data. Table 1 gives the comparison of each method with the ground truth, while Table 2 shows the total memory use and runtime for each method on each dataset.

`SCEPTRE` is not quite the best in terms of these metrics, but is also never far from it. `CLEANSER` and `pertpy` both have the potential to fail. `SCEPTRE` strongly wins in terms of computational metrics.

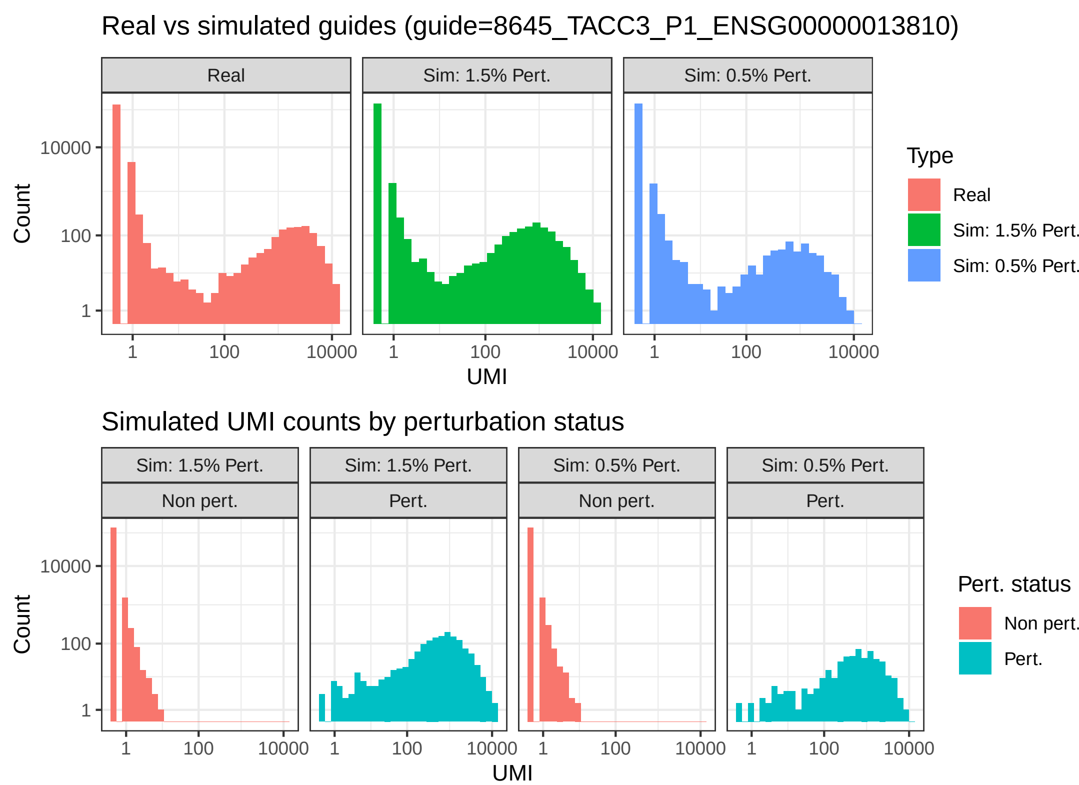

Figure 1: A comparison of the real guide from Replogle-RD7 that served as the basis for the simulations, as well as one guide from each of the specific simulated datasets. The first row shows that the simulated guides marginally match the real one. The second row shows the true components of each guide. The components are fairly well separated in both cases, as this is a high signal strength simulation.

Table 1: Metric means and (2.5%, 97.5%) quantiles over guides. Sim 1.5% has 25 guides, and Sim. 0.5% has 10 guides. Both datasets have 100k cells, so the expected number of actual perturbed cells are 1500 and 500, respectively.

Dataset	Method	Jaccard	Precision	Recall	Num. Assn. Cells
1.5% Perturbed	SCEPTRE	0.96 (0.94, 0.97)	0.98 (0.96, 0.99)	0.98 (0.98, 0.99)	1467 (1400, 1544)
1.5% Perturbed	crispat	**0.96 (0.95, 0.97)**	**1.00 (0.99, 1.00)**	0.96 (0.95, 0.97)	1403 (1357, 1469)
1.5% Perturbed	pertpy	0.08 (0.00, 0.81)	0.26 (0.00, 1.00)	0.08 (0.00, 0.81)	1013 (0, 5544)
1.5% Perturbed	CLEANSER	0.43 (0.42, 0.45)	0.43 (0.42, 0.45)	**1.00 (0.99, 1.00)**	3354 (3230, 3447)
0.5% Perturbed	SCEPTRE	0.93 (0.89, 0.96)	0.96 (0.90, 0.99)	0.97 (0.95, 0.98)	538 (488, 596)
0.5% Perturbed	crispat	**0.95 (0.93, 0.96)**	**0.98 (0.97, 0.99)**	0.97 (0.95, 0.98)	528 (497, 563)
0.5% Perturbed	pertpy	0.11 (0.00, 0.75)	0.51 (0.01, 0.99)	0.11 (0.00, 0.75)	258 (0, 1537)
0.5% Perturbed	CLEANSER	0.22 (0.21, 0.23)	0.22 (0.21, 0.23)	**1.00 (0.99, 1.00)**	2454 (2401, 2531)

Table 2: Simulations: Computational Performance

Method	Runtime (min)	Peak memory (GB)
0.5% Pert (10 guides)		
SCEPTRE	**0.21**	**0.31**
crispat	8.70	1.10
CLEANSER	13.80	2.90
pertpy	7.77	3.90
1.5% Pert (25 guides)		
SCEPTRE	**0.37**	**0.31**
crispat	10.53	1.10
CLEANSER	50.18	8.60
pertpy	12.98	5.40

2.1.2 Results on real data

The downsampled Replogle and Gasperini datasets are used here, so they are denoted `medium`. Both have 50k cells and 2500 guides. Figure 2 shows the agreement between each pair of methods on each guide. All but `pertpy` generally agree. Table 3 shows the estimated experimental MOI and the average number of cells assigned to each guide. We again see `pertpy` as an outlier. Table 4 shows the computational results for these 4 methods, as well as for the `sceptre-pipeline`. `SCEPTRE` and the pipeline are by far the fastest and most memory efficient.

Overall we see a picture of `SCEPTRE` being nearly the best method statistically, and by far the best computationally, so for large-scale guide assignment it is a very strong choice.

2.2 Association: positive control analysis

Next, we turn to association analysis. For this part of the project, we are currently comparing 3 methods on 2 real datasets.

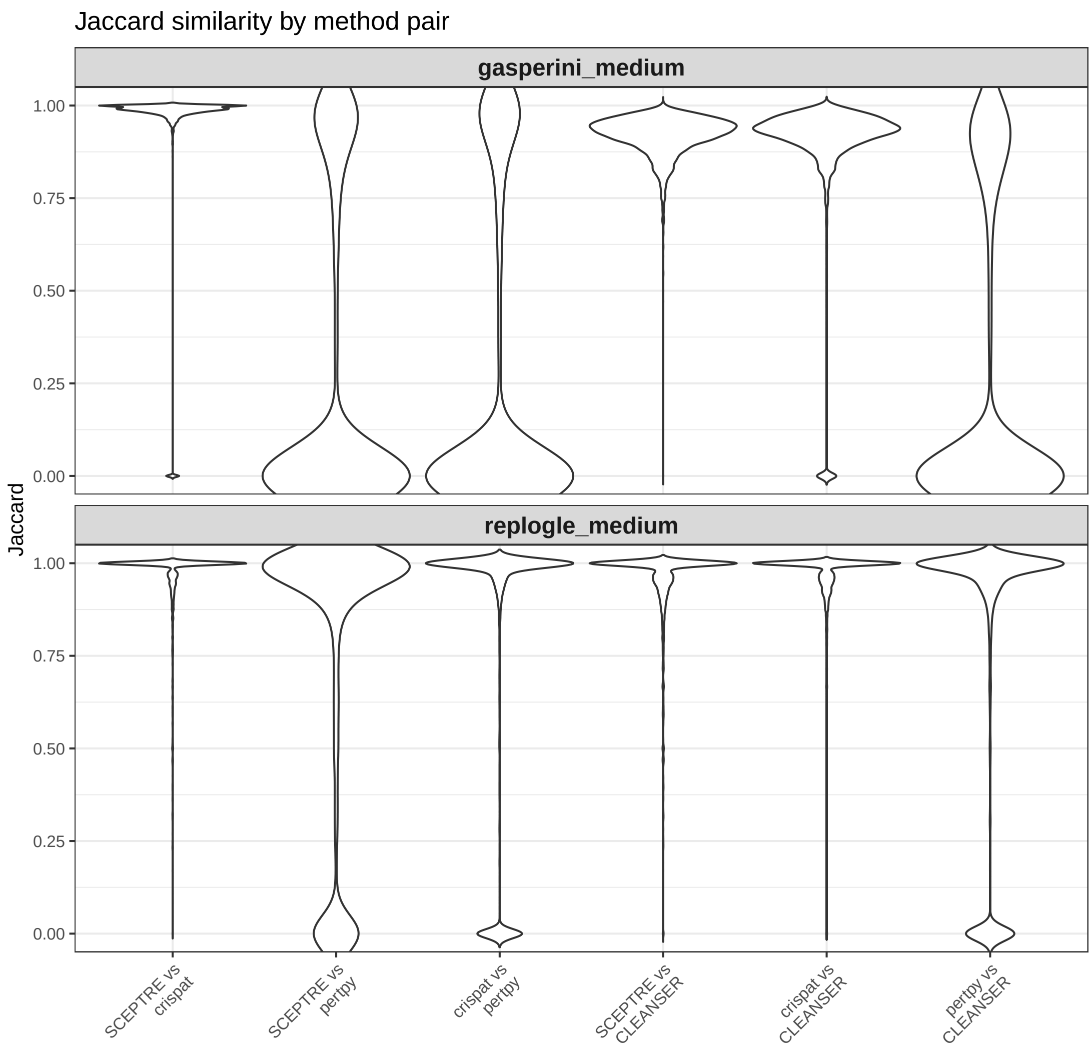

Figure 2: A comparison of the Jaccard similarities between the assignments made for each guide by each method pair. We can see that `SCEPTRE`, `crispat`, and `CLEANSER` generally agree, but `pertpy` can be quite different.

Table 3: Average assignment sizes: cells per gRNA and gRNAs per cell.

Dataset	Method	Avg cells per gRNA	Avg gRNAs per cell
gasperini_medium	SCEPTRE	131.9	6.68
	crispat	129.1	6.57
	pertpy	90.6	4.73
	CLEANSER	122.5	6.29
replogle_medium	SCEPTRE	27.0	1.54
	crispat	25.3	1.56
	pertpy	33.9	2.07
	CLEANSER	24.5	1.54

Table 4: Real data: Computational Performance. For `sceptre-pipeline`, the peak memory is the maximum from a single node. The average number of concurrent workers for `sceptre-pipeline` was 31 and 14 for Gasperini and Replogle-RD7, respectively.

Method	Runtime (min)	Peak memory (GB)
gasperini_medium		
SCEPTRE	**10.63**	**0.29**
crispat	103.03	1.40
CLEANSER	217.55	54.10
pertpy	533.00	7.70
replogle-rd7_medium		
SCEPTRE	**12.43**	**0.30**
crispat	107.92	1.40
CLEANSER	211.62	49.60
pertpy	660.48	8.40
Replogle-RD7		
sceptre-pipeline	13.4	0.744
Gasperini		
sceptre-pipeline	11.4	0.644

The methods are as follows:

1. `FR-Perturb`: An approach based on sparse factorization. Both the resampling and skew-t variants are included.

2. `Mixscale`: Inference is done via a NB regression using Mixscape-style scores as weights to account for perturbation effectiveness. Low MOI only.

3. `SCEPTRE`

The datasets are as follows:

1. **Replogle-RD7 on-target**: the Replogle-RD7 data, downsampled to around 210k cells, with around 1600 on-target genes, corresponding to around 1800 guides. Only on-target genes were included where every guide for them had at least 20 cells expressing it. Cell expressions were obtained via the `sceptre-pipeline`.

2. **Gasperini on-target**: the Gasperini data, downsampled to around 185k cells, with almost 400 on-target genes, corresponding to around 850 guides. Cell expressions were obtained via the `sceptre-pipeline`.

Table 5 gives some results for each dataset-method pair. SCEPTRE has by far the smallest p-values. Next, Table 6 shows that SCEPTRE and Mixscale give very similar rankings to the on-target genes, and the two FR-Perturb methods strongly agree with each other as well, but there is very little agreement between FR-Perturb and the other methods. Finally, Figure 3 shows the distribution of the $-\log_{10}$ p-values for each method. FR-Perturb can be seen to be much more concentrated on large p-values.

Table 5: Summary statistics by dataset and method. The column 'Corr. with n' gives the Spearman correlation of the negative p-value with the number of cells expressing each target. 'Pcnt. ≤ 0.05' gives the percentage of p-values at or below 0.05. Both datasets are downsampled to only have on-targets. `SCEPTRE` has by far the smallest on-target p-values.

Dataset	Method	Num. Targets	Mean p-value	Corr. with n	Pcnt. ≤ 0.05
Gasperini	FR-Perturb (resamp)	377	0.2510	0.031	43%
Gasperini	FR-Perturb (skew)	377	0.2530	0.022	43.5%
Gasperini	SCEPTRE	377	0.0113	0.177	98.1%
Replogle	FR-Perturb (resamp)	1619	0.2850	0.068	33.5%
Replogle	FR-Perturb (skew)	1619	0.2840	0.067	34.1%
Replogle	Mixscale	1619	0.1650	0.313	78.1%
Replogle	SCEPTRE	1619	0.0456	0.318	87.7%

3 Next Steps

Below I outline all of the remaining TODOs I have for this project, along with my best guess for time estimates.

I believe I can complete nearly all of the Assignment and Association TODOs by the first week of March, leaving over a month for the final writing stage.

1. General

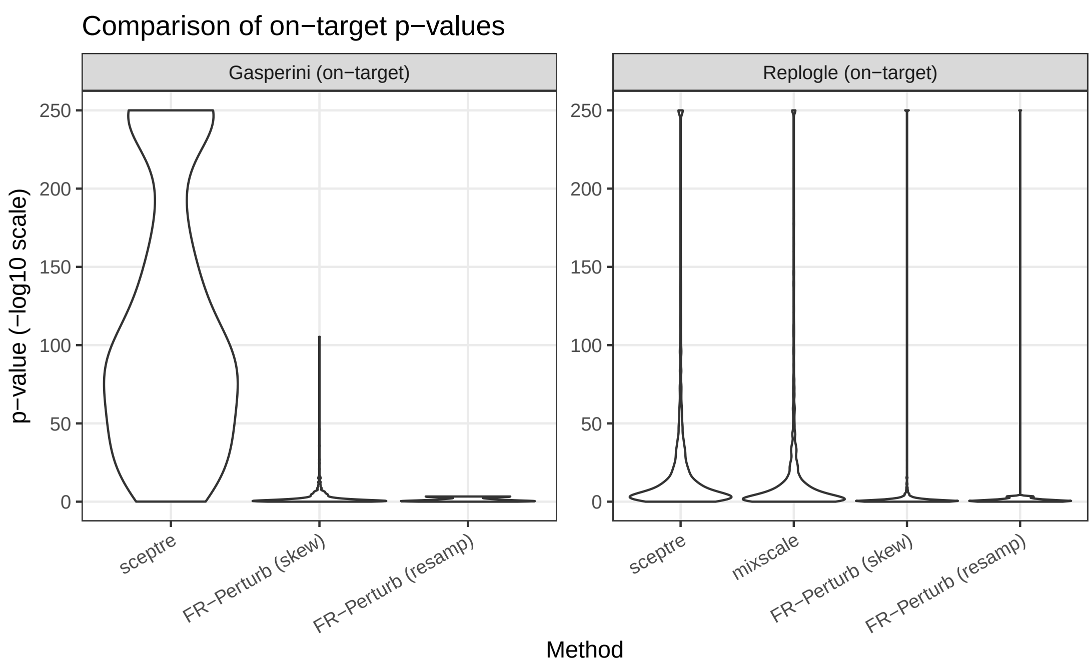

Figure 3: A comparison of the -log10 p-values for each method. FR-Perturb assigns much larger p-values in general.

Table 6: Pairwise method comparisons within each dataset. 'Pcnt. $1 \leq 2$' gives the percentage of times the p-value for Method 1 is at or below the p-value for Method 2. 'Corr.' gives the Spearman correlation between the p-values for Methods 1 and 2. Both datasets are downsampled to only have on-targets.

Dataset	Method 1	Method 2	Pcnt. $1 \leq 2$	Corr.
Gasperini	FR-Perturb (skew)	FR-Perturb (resamp)	59.4%	0.990
Gasperini	SCEPTRE	FR-Perturb (resamp)	98.4%	0.233
Gasperini	SCEPTRE	FR-Perturb (skew)	98.1%	0.238
Replogle	FR-Perturb (skew)	FR-Perturb (resamp)	55.8%	0.996
Replogle	Mixscale	FR-Perturb (resamp)	77.9%	0.007
Replogle	Mixscale	FR-Perturb (skew)	76%	0.005
Replogle	SCEPTRE	FR-Perturb (resamp)	89%	0.014
Replogle	SCEPTRE	FR-Perturb (skew)	86.8%	0.013
Replogle	SCEPTRE	Mixscale	81.1%	0.926

   (a) Switch to array jobs? [3 days]

2. Assignment
   (a) Add more simulated datasets: try the approach of fishash? [4 days]
   (b) Add more real datasets: barnyard in particular [4 days]
   (c) Add more methods: geomux in particular [4 days]
   (d) Add more assessments, perhaps using hold-outs or resampling? [low priority]
   (e) Incorporate positive and negative control analysis into guide assignment assessment [3 days]

3. Association
   (a) Implement and analyze negative control analysis [7 days]
   (b) Implement and analyze computational benchmarking [7 days]
   (c) For positive control analysis, consider more methods and more datasets [7 days]
   (d) For positive control analysis, consider using covariates to understand the results [1 day]
   (e) For positive control analysis, include effect size results [1 day]

4. Writing
   (a) The benchmarking paper [1 month?]
   (b) Thesis defense presentation [14 days]
   (c) Thesis [14 days?]